



Percentage increase and decrease

Teacher resource

Question 1

A chair usually costs \$360.
The chair is on sale for 5% off.

What is the sale price of the chair?

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| \$295 | \$342 | \$355 | \$365 |
| ☐ | ☐ | ☐ | ☐ |

Answer: B

Skill: Students calculate percentage decrease.

Background information

Students should be encouraged to develop a range of informal strategies to calculate and approximate percentages. For example, 25% can be calculated by halving and halving again, 15% can be thought of as 10% plus one half of 10%, 60% can be thought of as half plus 10%. Such strategies can be used for mental calculation or, if needed, recorded in writing as informal written calculations.

Additional questions

1. What is 25% of \$250?
2. Calculate 15% of \$80.
3. Evaluate 40% of \$150.

Question 2

Jade saved \$240 by the end of June.

This is 25% more than she had saved by the end of May.

How much had Jade saved by the end of May?

\$180
☐

\$190
☐

\$192
☐

\$288
☐

Answer: C

Skill: Students calculate using percentages.

Background information

Students may need support to determine what amount represents the whole or 100%.
In this example, \$240 represents 125% of the amount Jade had saved by the end of May.

Additional questions

How much would Jade have saved by the end of May if \$240 was:

1. 50% more than she had saved by the end of May?
2. 50% less than she had saved by the end of May?
3. 20% more than she had saved by the end of May?

Question 3

A frozen drink has a volume of 240 cm^3 .

After melting for 20 minutes the volume is 192 cm^3 .

By what percentage has the volume decreased?

10
☐

20
☐

25
☐

50
☐

Answer: C

Skill: Students express an increase or decrease of an amount as a percentage.

Additional questions

What percentage is \$40 of the following amounts?

1. \$50
2. \$200
3. \$30
4. \$4

Question 4

A woman had three savings accounts.

The table below shows the initial amount and percentage increase of each account for one year.

Starting amount (dollars)	Percentage increase
20 000	10
20 000	?
10 000	5

The percentage increase of her total savings was 20%.

Calculate the value of ? in the table.

Answer: 37.5

Skill: Students calculate with percentages.

Background information

Students may be able to calculate percentages of an amount as well as percentage increases and decreases between numbers but not understand under which circumstances it is reasonable to add and subtract percentages.

Additional questions

Calculate the value of ? in the table if the percentage increase in her total savings was:

1. 10
2. 50
3. 5

Student activity: Percentage increase and decrease

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